

CHAPTER 6

CONCLUSIONS AND FUTURE WORKS

6.1 Conclusion

This thesis set out to determine whether stating a manipulability safety requirement as an explicit constraint, rather than folding it into the reward, changes the safety behaviour of a learned grasping policy on a UR5e manipulator, and whether it changes how predictable that behaviour is across independent training runs (Section 1.3). Both questions are answered. Constraining the policy reduces unsafe behaviour, and its larger and less anticipated effect is on how consistently that behaviour is reproduced from one training run to the next (Section 4.10). The six specific objectives set out at the start of this thesis were each met in turn.

The first objective was to implement an unconstrained PPO baseline and a constrained PPO-Lagrangian agent for a UR5e pick-and-lift grasping task in Isaac Lab, trained under a shared floor on manipulability. Both were built and trained as specified, on the frozen weld environment described in Chapter 3, at the calibrated thresholds recorded in Section 4.1.

The second objective was to isolate the effect of the constraint itself from the effect of the extra cost critic a Lagrangian method carries, by training a control arm whose multiplier is pinned at zero throughout. That isolation succeeded more completely than the design anticipated: Section 4.2 found the control arm’s stored network weights identical to the unconstrained baseline’s, on every seed, so the artifact term in the decomposition of Equation (4.1) is zero by direct verification rather than by assumption, and every safety difference reported in this thesis can be attributed to the constraint alone.

The third objective was to establish whether the constrained agent’s behaviour depends on how tight its budget is, by training it at two episodic cost budgets, 25 and 15, rather than one. Section 4.8 answers it: tightening the budget continued to help within the range tested, reducing true singularity crossings further and leaving joint-limit contact at zero, at no measurable cost to task performance, though the worst single-instant manipulability was unmoved by the tighter budget as much as by the looser one.

The fourth objective was to train every arm on five independent seeds and report outcomes as a distribution rather than a single mean. That is the reporting convention followed throughout Chapter 4, and it is what exposed the finding the general objective was actually asking about: the unconstrained baseline’s safety outcome is not one number but a distribution spanning a factor of twenty, and no telemetry available during training would have told an engineer

which end of that distribution a given run had landed on.

The fifth objective was to evaluate every trained policy on frozen, held-out episodes, so the reported figures describe the policy that would actually be deployed rather than one still exploring. Section 4.1 sets out that protocol, forty-five thousand evaluation episodes across three held-out seeds, and it exists because an earlier version of this project’s evaluation had read safety statistics from a still-learning policy and reported a difference that later proved to be measuring an implementation artifact rather than the constraint (Section 4.2). The corrected protocol is what makes every other objective’s finding trustworthy.

The sixth and final objective was to determine whether the constraint changes the average safety outcome of a training run, its spread, or both. It changes both, but not by equal amounts. On the mean, the constrained agent reduces true singularity crossings by a factor of 7.0 at a budget of 25 and 39.9 at a budget of 15, and eliminates all observed joint-limit contact at both, against 10.70 % of baseline episodes touching one. On the spread, which Section 4.6 identifies as the larger effect, a twentyfold seed-to-seed range in episodic safety cost collapses to roughly two-to-fivefold, and the same collapse turns up unprompted in task precision, where the seed-to-seed standard deviation of sub-centimetre success falls from about 7.5 percentage points to about one. Neither effect cost the constrained agent anything on the task itself: reward and goal-reach are equal to or better than the baseline’s at both budgets.

Read together, the six objectives support the general objective’s answer directly. A stated safety budget does change what a learned grasping policy does, and its main effect is not a modest average improvement but the near-elimination of a lottery in which an unconstrained training run might, by the luck of its random seed alone, produce a policy that touches a joint limit on nearly half its episodes. Whether that policy is the one shipped to hardware is not something an unconstrained training procedure lets an engineer know in advance. A constrained one does.

6.2 Future Works

Five extensions follow from what this thesis chose not to attempt or could not finish, each positioned here rather than apologised for.

The most direct is closing the visual loop. This thesis trained on privileged object pose, the cube’s true position handed to the policy directly, because isolating the safety comparison from a perception problem was the point of Layer 1 (Section 1.4). The registered programme’s second layer would replace that privileged pose with an eye-in-hand RGB-D camera and object detector, and use the learned policy to tune the image Jacobian relating pixel motion to joint motion rather than fixing it in advance, extending the fixed, hand-derived visual servoing loop of the predecessor project in this laboratory [?] into a learned, adaptive one.

Whether a constraint framed on manipulability continues to hold once the arm is also reacting to a noisy, partially occluded camera feed is not something this thesis’s simulation-only, full-state-observation result can answer, and is the natural next question.

The second is sim-to-real transfer to the physical UR5e, the programme’s third layer. The trained policy already exports to both JIT and ONNX from the same checkpoint format used throughout this thesis, so the software path to a ROS 2 Humble deployment exists. What does not yet exist is a resolution to the largest known transfer gap: the simulation grasps with a lumped-mass abstraction of a Robotiq 2F-85, welding the cube to the tool frame at the moment of contact rather than modelling the grip physically, while the laboratory’s real arm carries a ROBOTIS RH-P12-RN, a different gripper with different kinematics. Closing that gap is gripper-specific engineering, not a parameter to randomise away, and it is Layer 3’s opening task rather than an afterthought to it.

The third is the comparison this thesis registered but did not complete. The plan called for an off-policy arm, soft actor-critic, alongside the on-policy PPO and PPO-Lagrangian agents actually trained; it was cut on schedule grounds, not because a reason emerged to expect it would behave differently. Continuous-control grasping has been demonstrated with both algorithm families [?], so nothing here argues against completing the comparison, only that time ran out before it could be run under the same fairness protocol as the rest of this batch.

The fourth is a specific and already-evidenced improvement to the constrained optimiser itself. Section 4.7 measured the Lagrange multiplier overshooting on every seed at both budgets before settling, a single sharp rise peaking between iterations 47 and 58 followed by a decay to zero. That is the textbook signature plain dual ascent is known to produce, integral control with nothing to damp it, and it is exactly the problem Stooke *et al.* [?] built PID-Lagrangian to correct, by adding proportional and derivative terms to the multiplier update. Because this thesis measured the overshoot rather than merely suspecting it, PID-Lagrangian is not a speculative extension here. It is a direct response to a documented result in Section 4.7, and the natural successor to the plain dual-ascent update used throughout this work.

The fifth is scale in the one dimension this thesis deliberately kept small: the number of seeds. Five seeds were enough to show that unconstrained training on this task produces a wide and unpredictable range of safety outcomes and that the constraint narrows it (Section 4.6), because the effect is large enough to appear clearly even at that sample size. Five seeds are not enough to attach a confidence interval to how large the narrowing is, a limitation Section 4.9 states directly. A larger seed count, beyond what this thesis’s schedule allowed, would turn a demonstrated effect into a quantified one.